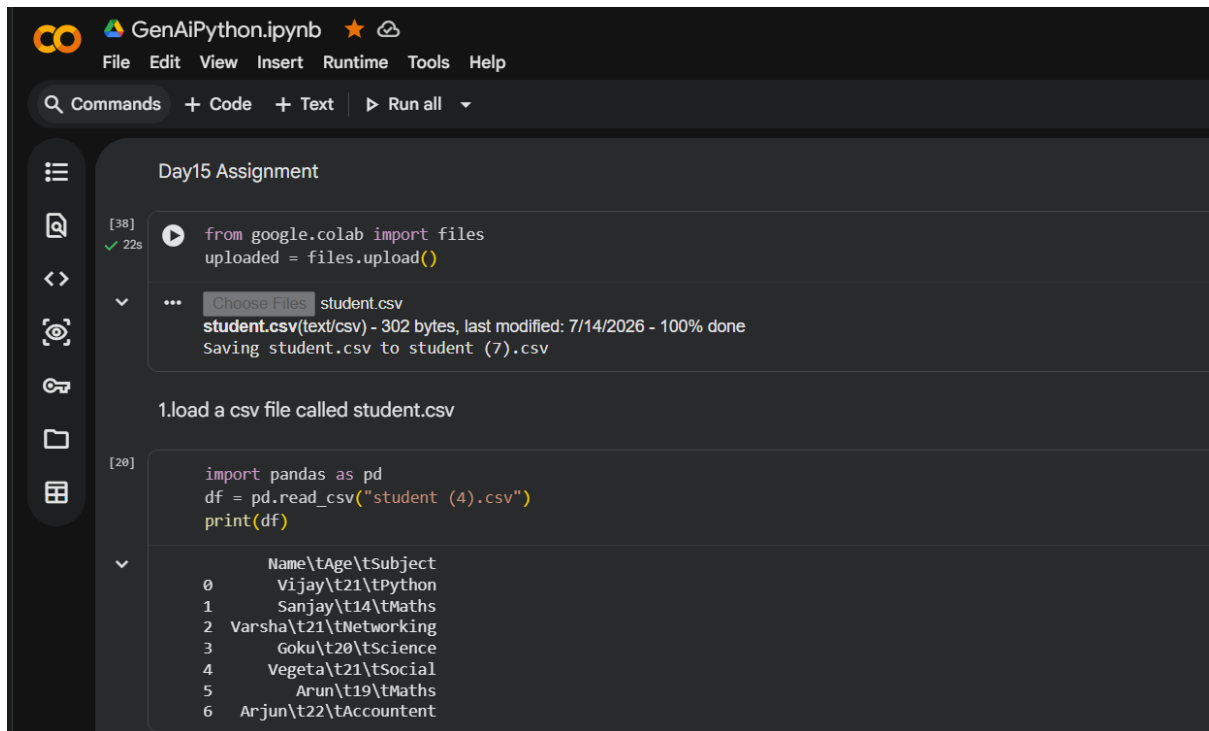


Day 15 Assignment

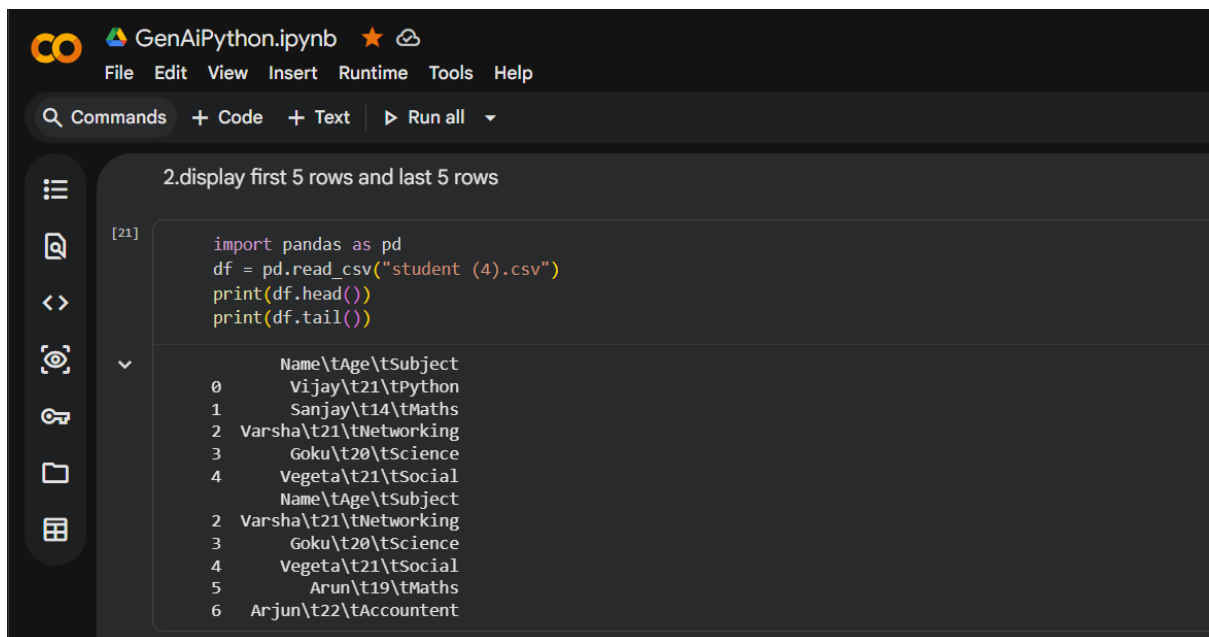


The screenshot shows a Jupyter Notebook titled "GenAiPython.ipynb". The interface includes a menu bar (File, Edit, View, Insert, Runtime, Tools, Help) and a toolbar with options like "Commands", "Code", "Text", and "Run all". The notebook content is titled "Day15 Assignment".

Cell [38] shows the upload of a file named "student.csv" from Google Colab. The output indicates the file is 302 bytes, last modified on 7/14/2026, and is 100% done.

Cell [20] shows the loading of the CSV file into a pandas DataFrame. The output displays the first 7 rows of the data:

	Name	Age	Subject
0	Vijay	21	Python
1	Sanjay	14	Maths
2	Varsha	21	Networking
3	Goku	20	Science
4	Vegeta	21	Social
5	Arun	19	Maths
6	Arjun	22	Accountant



The screenshot shows the same Jupyter Notebook interface. Cell [21] shows the code to display the first 5 rows and last 5 rows of the DataFrame. The output shows the first 5 rows and the last 5 rows of the data:

	Name	Age	Subject
0	Vijay	21	Python
1	Sanjay	14	Maths
2	Varsha	21	Networking
3	Goku	20	Science
4	Vegeta	21	Social

	Name	Age	Subject
2	Varsha	21	Networking
3	Goku	20	Science
4	Vegeta	21	Social
5	Arun	19	Maths
6	Arjun	22	Accountant

GenAiPython.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

3.find no of rows and no of columns

```
[22] import pandas as pd
df = pd.read_csv("student (4).csv")
print(df.shape)
```

(7, 1)

4.display all columns names

```
[23] import pandas as pd
df = pd.read_csv("student (4).csv")
print(df.columns)
```

Index(['Name\tAge\tSubject'], dtype='object')

5.find the datatype of every columns

```
[24] import pandas as pd
df = pd.read_csv("student (4).csv")
print(df.dtypes)
```

Name\tAge\tSubject object
dtype: object

GenAiPython.ipynb

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6.generate the statistical summary

```
[25] import pandas as pd
df = pd.read_csv("student (4).csv")
print(df.describe())
```

Name\tAge\tSubject
count 7
unique 7
top Vijay\t21\tPython
freq 1

7.display three random rows

```
[26] import pandas as pd
df = pd.read_csv("student (4).csv")
print(df.sample(3))
```

...
4 Vegeta\t21\tSocial
1 Sanjay\t14\tMaths
2 Varsha\t21\tNetworking

GenAiPython.ipynb

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8.find how many students belong to each city using value_counts

```
[32] import pandas as pd
df = pd.read_csv("student (4).csv")
print(df["Subject"].value_counts())
```

Subject
Maths 2
Python 1
Networking 1
Science 1
Social 1
Accountant 1
Name: count, dtype: int64

GenAiPython.ipynb

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9. display only the name column

```
[34] import pandas as pd
df = pd.read_csv("student (5).csv")
print(df["Name"])
```

0 Vijay
1 Sanjay
2 Varsha
3 Goku
4 Vegeta
5 Arun
6 Arjun
Name: Name, dtype: object

GenAiPython.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

10. display name and salary columns

```
[35] import pandas as pd
df = pd.read_csv("student (5).csv")
print(df[["Name", "Salary"]])
```

	Name	Salary
0	Vijay	400000
1	Sanjay	350000
2	Varsha	300000
3	Goku	100000
4	Vegeta	40000
5	Arun	100000
6	Arjun	200000

11. display students whose marks are greater than 80

```
[37] import pandas as pd
df = pd.read_csv("student (6).csv")
print(df[df["Marks"]>80])
```

	Name	Age	Subject	Salary	Marks
0	Vijay	21	Python	400000	90
1	Sanjay	14	Maths	350000	85
2	Varsha	21	Networking	300000	95
5	Arun	19	Maths	100000	88

GenAiPython.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

12. display employee from chennai

```
[39] import pandas as pd
df = pd.read_csv("student (7).csv")
print(df[df["Location"]=="Chennai"])
```

	Name	Age	Subject	Salary	Marks	Location
0	Vijay	21	Python	400000	90	Chennai
2	Varsha	21	Networking	300000	95	Chennai

13. Sort the dataset based on salary in descending order

```
[46] import pandas as pd
df = pd.read_csv("student (7).csv")
print(df.sort_index(ascending=False))
```

	Name	Age	Subject	Salary	Marks	Location
6	Arjun	22	Accountant	200000	78	Bangalore
5	Arun	19	Maths	100000	88	Cuddalore
4	Vegeta	21	Social	40000	77	Bangalore
3	Goku	20	Science	100000	60	Bangalore
2	Varsha	21	Networking	300000	95	Chennai
1	Sanjay	14	Maths	350000	85	Cuddalore
0	Vijay	21	Python	400000	90	Chennai

GenAiPython.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

14. add a bonus column equal to 15% of salary

```
[63] import pandas as pd
df = pd.read_csv("student (7).csv")
df["Bonus"] = df["Salary"] * 0.15
df.to_csv("student (7).csv", index=False)
print(df)
```

	Name	Age	Subject	Salary	Marks	Location	Bonus
0	Vijay	21	Python	400000	90	Chennai	60000.0
1	Sanjay	14	Maths	350000	85	Cuddalore	52500.0
2	Varsha	21	Networking	300000	95	Chennai	45000.0
3	Goku	20	Science	100000	60	Bangalore	15000.0
4	Vegeta	21	Social	40000	77	Bangalore	6000.0
5	Arun	19	Maths	100000	88	Cuddalore	15000.0
6	Arjun	22	Accountant	200000	78	Bangalore	30000.0

GenAiPython.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

15. rename the column name to employee_name

```
[64] import pandas as pd
df = pd.read_csv("student (7).csv")
df.rename(columns={"Name": "employee_name"}, inplace=True)
print(df)
```

	employee_name	Age	Subject	Salary	Marks	Location	Bonus
0	Vijay	21	Python	400000	90	Chennai	60000.0
1	Sanjay	14	Maths	350000	85	Cuddalore	52500.0
2	Varsha	21	Networking	300000	95	Chennai	45000.0
3	Goku	20	Science	100000	60	Bangalore	15000.0
4	Vegeta	21	Social	40000	77	Bangalore	6000.0
5	Arun	19	Maths	100000	88	Cuddalore	15000.0
6	Arjun	22	Accountant	200000	78	Bangalore	30000.0

GenAIPython.ipynb

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16.delete the bouns column

```
[66] import pandas as pd
      df = pd.read_csv("student (7).csv")
      print(df.drop("Bonus",axis=1))
```

	Name	Age	Subject	Salary	Marks	Location
0	Vijay	21	Python	400000	90	Chennai
1	Sanjay	14	Maths	350000	85	Cuddalore
2	Varsha	21	Networking	300000	95	Chennai
3	Goku	20	Science	100000	60	Bangalore
4	Vegeta	21	Social	400000	77	Bangalore
5	Arun	19	Maths	100000	88	Cuddalore
6	Ar-jun	22	Accountent	200000	78	Bangalore